Operations and Installation Instructions

2315 Archiver on Diablo 31 Drive

Archiving 2315 cartridges for the IBM 1130 System

The IBM 1130 System uses a single 14” platter disk cartridge (2315) inserted into an internal disk drive (13SD or Ramkit) or the optional 2310 external disk drive cabinet.

The 2315 cartridges record up to 512K words in sectors of 321 words each. The disk is organized into 203 cylinders, with an upper and a lower surface (track) per cylinder and having four sectors on each track. The platter is inside a protective plastic case.

The data is recorded at a rate of 720 kilobits per second while the disk platter spins at 1500 rpm. Thus data streams to and from the drive at 36K words per second during a read or write.

Each 16 bit word is recorded with four additional error checking bits. There is a mandatory preamble and a sync word at the start of each sector. The format for a sector is unique to the IBM 1130 and 1800 systems.

IBM licensed disk drive technology to multiple companies

IBM offered licenses to other manufacturers who produced disk drives and disk cartridges based on the 13SD drive and 2315 cartridge designs. Several firms made disk cartridges and several developed disk drives for their minicomputer systems under license.

The 2315 cartridge used with the IBM 1130 has eight physical sectors around the circumference of the platter, but licensees made versions with other sector counts, up to 32. The platter is otherwise identical except for the metal hub with its sector mark notches.

The companies that licensed the disk technology were free to improve on the 13SD drive and most did. The IBM drive used a mechanical ratchet system for arm movement between cylinders, thus is moved in steps of 1 or 2 cylinders at a time. All the other manufacturers designed a servo system that supports a single move of any number of cylinders.

The rate at which data is clocked onto the platter - 720KHz in the case of the IBM drive - was implemented at higher rates of 781, 1440 and 1562 KHz by those makers. The two higher rates were labeled High Density.

Other versions doubled the number of cylinders on a pack by writing narrower tracks and moving in finer increments. A few added a second fixed platter underneath the place where the cartridge's platter spins. Thus the licensees could achieve more than 16 times the data density of the 13SD and 2315 used in IBM 1130 systems.

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Diablo Model 31 Drives

Diablo Systems built their Model 30 series under license and in particular sold the Model 31 as the closest version to the IBM 13SD/2310. They offered many options - Model 31 could operate at any of the four data rates and make use of cartridges with 8, 12, 16, 24 or 32 sectors per revolution. Xerox PARC (Palo Alto Research Center) created their groundbreaking Alto computers using a high density model 31 drive.

Model 31 drives can be used with this project to read 2315 cartridges that were written on IBM 1130 systems, but must match the specifications of the 13SD drive. Thus, they must use 8 sector cartridges and have a data rate of 720KHz. If so, the Diablo drive and the 1130 can read data written on the other system, giving full interchangeability.

The Model 31 must have certain options listed to be compatible with the 13SD. Essentially, they must be a standard density version, at either 720KHz or 781KHz data rate.

Modifications to the printed circuit boards inside the Diablo drive can change a 781KHz drive into the desired 720Khz for compatibility, but option 001 (720KHz) works without modification. Details of the modification are provided later in this document.

Options26 39, 40, 41, 42, 51, 52, or 54 indicate drives that are based on more sectors than the 8 per revolution used with the IBM drives. The other options may slightly change the electrical interface but don't affect the physical interchangeability of cartrdiges between IBM and Diablo.

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This is the full option list for Model 31 drives:

Option	Description
001	720 KHz – 1020BPI – IBM Compatible
002	781 KHz – 1100BPI – Diablo Standard
003	1440 KHz – 2040BPI – Option
004	1562 KHz – 2200BPI – Option
005	Write Protect
008	Erase Gate (Not tied to Write Gate)
013	Paint – Orange with Diablo Logo
014	Paint – Orange without logo
015	Unpainted without logo
017	Flangeless Mount (tabletop trim)
018	Flush Mount
019	Extended Mount
020	Upper and Lower Trim for Flush Mount
024	Extender Board Assembly
025	Standard Density – 8 Sector
026	High Density – 8 Sector
027	Standard Density Alignment Pack
029	Dual Unit Power Supply
030	High Density Alignment Pack
032	Power Supply Cable
033-05′	Receptacle Connectors on 5′ flat cable (female)
033-07′	Receptacle Connectors on 7′ flat cable (female)
034-05′	Pin Connectors on 5′ flat cable (male)
034-07′	Pin Connectors on 7′ flat cable (male)
035	Receptacle terminator with resistors (female)
036	Pin terminator with resistors (male)
037	Receptacle terminator without resistors (female)
038	Pin terminator without resistors (male)
039	Standard Density – 12 Sector
040	Standard Density – 24 Sector
041	High Density – 12 Sector
042	High Density – 24 Sector
044	Rack Mounting for Power Supply
045	Paint – Black without logo
046	Attention Line Negative
047	Attention Line Positive
048	Interrupt Option
050	DTC Card FPC Interdata Controller (no longer available)
051	Standard Density – 16 Sector
052	High Density – 16 Sector
053-05′	M/F Cable 5′
053-07′	M/F Cable 7′
054	32 Sector
056	Cable Assembly (Interdata Controller)
057	Terminator (no erase gate) Female with resistors
058	Terminator (no erase gate) Male with resistors
060	Paint – Grey/Green without logo
061	Paint – Blue without logo
062	Grained without logo
063	Paint – Off White (Varian)
064	Attention line with logical address interlock (Varian)
065	Cartridge Closing Bail Assembly

Interface to Model 31 drive

The electrical interface to the Diablo model 31 differs from the IBM 13SD drive, mainly due to the change in arm movement between the IBM drive and the way that licensees like DEC, HP and Diablo implemented seeks.

The handshakes between the controller logic and the disk drive are different as well as having an absolute cylinder number and different movement command signals. Thus this cannot use the logic designed to read or write on the 13SD drives.

The Diablo drives can be daisy chained with up to four drives connected to a single controller, making use of four Select signals to control which drive is active at any instant. A terminator is installed on the back of the last drive in a chain. Terminators vary based on some options, thus when using this project, the builder must verify that the critical signal lines have termination resistors; the terminator must be modified if it does not match.

This project includes a printed circuit board that supports the electrical requirements of the interface to the Diablo Model 31, including termination resistors for the controller side of the cable. A standard 50 pin IDC connector (e.g an internal SCSI cable for a PC) is used to connect a cable from the PCB to the connection block of the Diablo Model 31 disk drive. This requires changes to the cable and connection of the wires in the cable to the proper pins on the disk drive.

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Components of the archiver project

A suitable Diablo Model 31 disk drive is required. A terminator and a signal connector are also needed, although the connector will be modified in all cases.

A Digilent Artix A7 FPGA board with the bitstream from this project loaded onto the board to implement the logic of the project.

A custom PCB installed on top of the FPGA board to provide the electrical interface between the FPGA and the Diablo drive.

A modified 50 pin IDC flat ribbon cable runs between the PCB and the Diablo drive connector.

A USB cable to provide a serial link to a PC, Mac or other computer where the data extracted from the 2315 cartridge can be uploaded.

A Python application that connects to the archiver and creates disk files in the format used by the IBM 1130 simulator programs. This is the standard for interchange of disk cartridge images.

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Operating the Archiver

With the disk drive and the power supply of the Archiver both plugged into the building AC mains, the device is ready to perform archiving.

It is highly recommended to open each 2315 cartridge before reading it, inspecting for damage, and cleaning the surfaces with 91% Isopropyl Alcohol and low/no lint wipes. This minimizes the risk that you will suffer a head crash which scapes up the disk heads in the Diablo drive and perhaps damages the 2315 cartridge platter.

Eight screws around the bottom of the cartridge are removed to open the cartridge case and expose the platter inside. Use disposable gloves when handling the platter as a fingerprint is high enough to cause a head crash if you touch the surface of the platter

The heads should also be inspected with a dental mirror before archiving each cartridge, looking for the buildup of oxide or signs of a head crash. Low lint swabs with 91% Isopropyl Alcohol can be used to clean the heads, although with crashes it may be necessary to remove the heads for a more complete cleaning and then perform an alighment on reinstallation.

Insert the cartridge into the Diablo drive and switch on the drive. I recommend repeatedly spinning the drive for about 30 seconds and shutting down to ensure all dust is removed first. The drive has a HEPA quality filter that should clean the air going into the cartridge, but it may take some time to get all residual dust out before we want to load the heads.

The cartridge should come up to speed and after 50 seconds, the Ready lamp on the drive illuminates indicating that the heads have been lowered onto the spinning disk surface.

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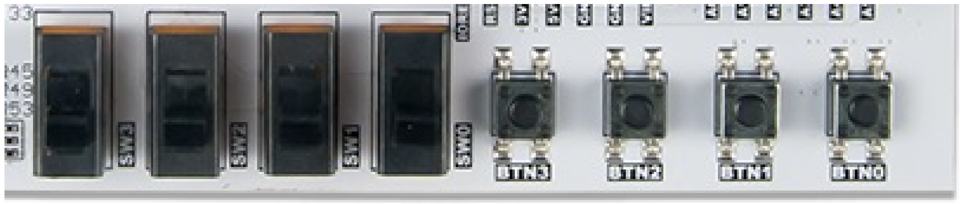


With the righrmost slide switch SW0 on the FPGA board set to the lower position, push the rightmost button BTN0 to begin the archiving process. LEDs will light with various status and error conditions - see the troubleshooting section later for details.

You will hear the Diablo drive moving the disk arm cylinder by cylinder as it reads all 203 cylinders into memory on the board. When it is complete, power down the Diablo drive and remove the 2315 cartridge.

Start the python application, choose a file location and name for the archived disk image, and wait until the application tells you it is ready.

Move slide switch SW0 to its top position and then push the button BTN0 to start the upload process. When the contents have been successfully loaded, the application will indicate success and you can move the rightmost slide switch back to the bottom position to prepare for the next cartridge to be archived.

The application will report any sectors that experienced data errors during the archiving. Since cylinders 200, 201, 202 and 203 are reserved as alternate cylinders for any of the first 200 that might have imperfections, you may see errors on some cylinders that were either defective or alternates that were not used nor formatted.

Note any errors just in case some content of the cartridge was not recovered properly. Deal with the error sectors manually using an IBM 1130 simulator and disk utilities, after first excluding defective and unused cylinders.

Building the cables between the archiver PCB and the Diablo drive

On the rear of the Diablo drive are a pair Winchester MRAC 42 plug and socket connectors and a Winchester MRAC-14S socket. The 42 pin parts are for signals while the 14 pin part delivers power.

Power connector

The power connector has letters A to F, H, J to N, P and R arranged left to right, from top to bottom looking at the face of the connector. Connect ground to pin C, high current (4A average 8A peak) +15V power supply to K, high current (4A average 8A peak) -15V power supply to R, low current +15V supply to H and low current -15V power supply wire to P. Use 16 ga wire for power and at least 15ga braid ground wire.

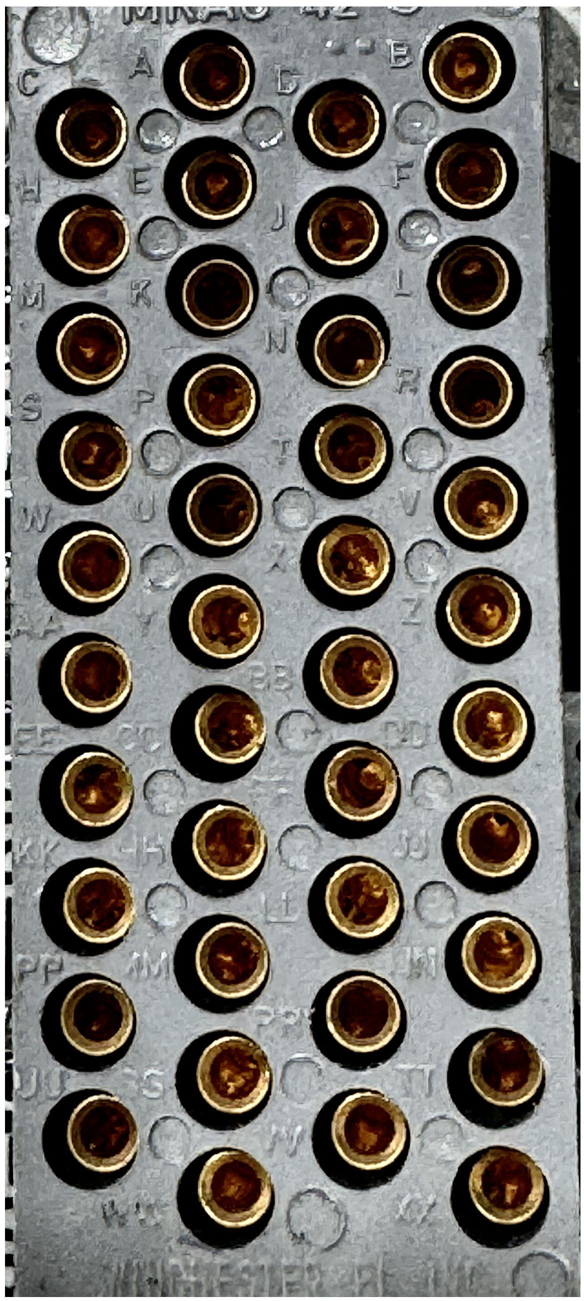
Noise generated by motors and servos will be fed back on the high current supplies, which is why a separate low current version is needed for signaling and logic.

Signal connector

There is a 42 position male (pins) and a female (sockets) connector on each drive. The terminator assembly is connected to one of them and the cable to the archiver connects to the other. The corresponding pins of both connectors are wired together thus the only difference is gender.

The connectors can be found on eBay, both the 42 pin shell and the male or female inserts for each position. Take a 50 cm 50 wire flat ribbon cable that has the IDC (internal SCSI) connectors installed. Remove on of the two IDC connectors and separate the wires. Every other wire in the cable is connected to ground on the archiver PCB.

The positions on the MRAC connector are labeled with single or double letter codes - A to F, H, J to N, P, R to Z, AA to FF, HH, JJ to NN, PP, and RR to XX running left to right, from top to bottom as you look at the male connector face.



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Connect wires 1 and 3 to pin D, wires 23 and 25 to pin DD and wires 47 and 49 to pin WW which establishes the ground connection to the drive. Then connect the following wires to these pins:

```
2  - NN
4  - SS
6  - AA
8  - Y
10 - E
12 - LL
14 - W
16 - N
18 - RR
20 - XX *
22 - J
24 - A
26 - X
28 - C
30 - FF
32 - T
34 - TT
36 - BB
38 - F
40 - VV
42 - EE
44 - B
46 - H *
48 - K *
50 - L
```

Wires 46 and 48 are attached only if the write protect (option 005) and erase gate (option 008) are installed on this drive.

If options 046, 047 or 064 are installed, do not connect wires 20 and 46.

Power supply sourcing

The drive requires four separate power supplies, two at +15VDC and two at -15VDC. One of each is a high current version, delivering a steady 4A with temporary bursts as the motor starts to spin. The other two deliver 2A of current for the logic and analog circuits that need to be free from the noise and impedance effects of the heavy current loads.

I bought three supplies, two are high current single voltage and the third delivers both +15V and -15V at lower current. These connect to the 120VAC mains. I found these on Amazon.com and should have them in a few days so that I can spin up the drive.

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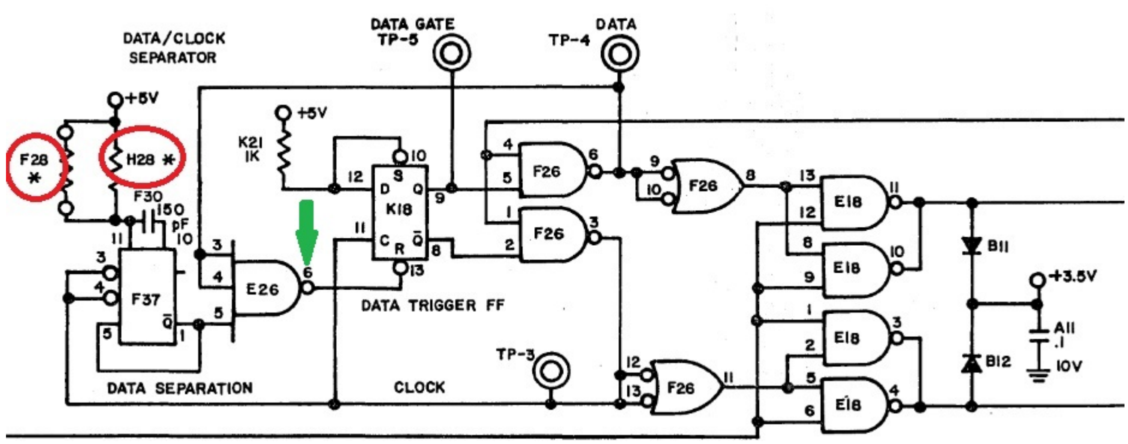
Modifying the Diablo Model 31 if option 002 (781KHz)

Pull out PCB J10 from the Diablo Model 31 drive's card cage. Verify it is the Standard Density version, not the 1440 or 1562KHz high density alternative which has a board number of 11082-xx.

The PCB board number will be 11113-04, 11113-05, 11113-06 or 11113-07 if it is the standard density 781KHz data rate board.If the PCB number is 11113-00, 11113-01, 11113-02 or 11113-03 then it is already the IBM compatible 720KHz design.

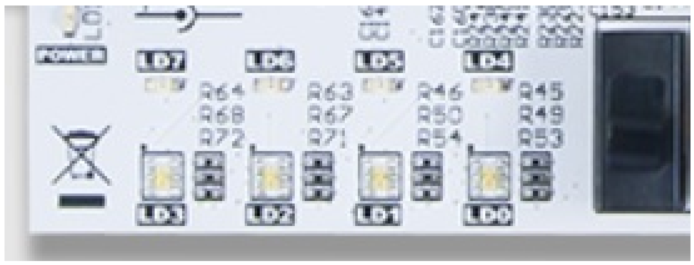
The four subtypes depend on if there is an independently controlled Erase Gate signal (option 008)and if it is designed for daisy-chaining up to four drives from the controller. The first and second are for machines with no daisy chain, which means no Selection jumper used; I have never seen one without daisy chaining. Thus you have the fourth type if option 008 is installed, otherwise the third type.

To convert the data rate down from 781KHz to 720KHz, you change the resistor at H28 from a 9.08K to a 10K 1/2W 5% part. You then trim the resistance by selecting a high value parallel resistor F28 so that the data gate signal, the output of gate E26, is high between 955 and 985 nanoseconds duration. The faster board had a shorter gate duration (882-912ns).



<u>Troubleshooting</u>

The FPGA board has eight LEDs that provide status information about the archiver operation - LD4 to LD7 are single color while LD0 to LD3 are tricolor and can show up to three signals per LED.



Displayed in both archive and upload

LD0 has red on head 1 is selected, off if head 0
LD0 has green on if the sector number is odd
LD0 has blue on if the sector number is 2 or 3

LD1 is blue if the arm is at cylinder 0 to 22
LD1 is green if the cylinder is 23 to 45
LD1 is red if the cylinder is 46 to 68

LD2 is blue if the cylinder is 69 to 91
LD2 is green if the cylinder is 92 to 114
LD2 is red if the cylinder is 115 to 147

LD3 is blue if the cylinder is 148 to 170
LD3 is green if the cylinder is 171 to 193
LD3 is red if the cylinder is 194 to 203

The signals displayed can vary based on the position of the rightmost slide switch SW0 where the down position is during archiving and the up position is during upload of the captured disk image.

Displayed during archiving

LD4 illuminates after power-on of the FPGA once the dynamic RAM on the board has finished its initialization and calibration. If this is not on, the archiver will not do anything.

LD5 is lit when the driver reports a logical address interlock - a seek attempt to a cylinder beyond 203. Most drives do not have this signal active so this would never light on those

LD6 is lit when a seek to a cylinder fails, which results in the drive becoming locked until it is power cycled. This is a hardware failure.

LD7 is lit when the archiving is complete - all 1, 632 sectors have been read and saved in RAM.

Displayed during uploading

LD4 illuminates when the FPGA has initialized and calibrated RAM so that it is ready to work. If this is not on, no other function works.

LD5 is lit when the upload process has completed, transferring all 1, 632 sectors of data over the USB serial link.

LD6 should not light in upload. It indicates that the disk drive suffered an error during a seek and has locked up, requiring power cycling.

LD7 is lit if the prior archiving process completed